ध्वसुधैव कुटुम्बकम्॥

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Subject: Programming with Java	**Sem: III AY: 2026-27**
Assignment No: 9	

TITLE: Write a Java program to demonstrate the use of the final keyword with variables, methods, and classes.

Problem Statement

Write a Java program to demonstrate the use of the final keyword with variables, methods, and classes.

Learning Objectives

To implement the final keyword with variables, methods, and classes.

Theory

The final keyword restricts modification in Java programs. A final variable cannot be reassigned after initialization. A final method cannot be overridden by subclasses, while a final class cannot be inherited. This improves security, prevents accidental modifications, and increases program reliability. Final is commonly used for constants and immutable classes.

Example code:

Users > sumit > Documents > java > FinalKeywordDemo.java > College

```java
final class College
{
    void displayCollege()
    {
        System.out.println(x: "College Name: SIT Pune");
    }
}

// Parent class
class Animal
{
    // Final method
    final void sound()
    {
        System.out.println(x: "Animals make different sound");
    }
}

// Child class
class Dog extends Animal
{
    void displayDog()
    {
        System.out.println(x: "Dog barks");
    }
```

```java
    // Uncommenting this will give a compilation error
    /*
    void sound()
    {
        System.out.println("Dog barks");
    }
    */
}
public class FinalKeywordDemo
{
    Run | Debug
    public static void main(String[] args)
    {
        // Final variable
        final int MAX_MARKS = 100;

        System.out.println("Maximum Marks = " + MAX_MARKS);

        Dog d = new Dog();
        d.sound();        // Inherited final method
        d.displayDog();

        College c = new College();
        c.displayCollege();
    }
}
```

Output:

```
java — -zsh — 161×46
Last login: Wed Jul 29 12:32:14 on ttys000
sumit@Sumit ~ % cd documents
sumit@Sumit documents % cd java
sumit@Sumit java % javac FinalKeywordDemo.java
sumit@Sumit java % ./FinalKeywordDemo
zsh: no such file or directory: ./FinalKeywordDemo
sumit@Sumit java % java FinalKeywordDemo
Maximum Marks = 100
Animals make different sound
Dog barks
College Name: SIT Pune
sumit@Sumit java % date
Thu Jul 30 11:45:33 IST 2026
sumit@Sumit java %
```

Exercise

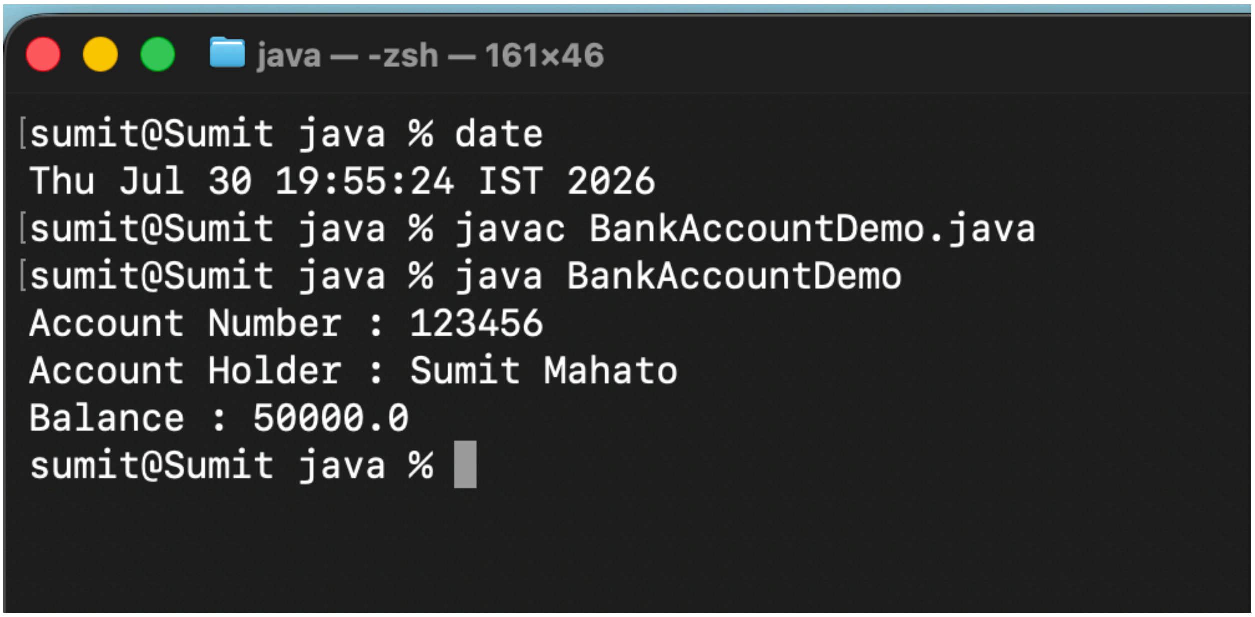
1. Create a **Bank Account program** where account number is final and cannot be changed.

Source Code:

Users > sumit > Documents > java > class BankAccount.java > BankAccountDemo

```java
class BankAccount
{
    final int accountNumber;
    String accountHolder;
    double balance;

    BankAccount(int accountNumber, String accountHolder, double balance)
    {
        this.accountNumber = accountNumber;
        this.accountHolder = accountHolder;
        this.balance = balance;
    }

    void display()
    {
        System.out.println("Account Number : " + accountNumber);
        System.out.println("Account Holder : " + accountHolder);
        System.out.println("Balance : " + balance);
    }
}

public class BankAccountDemo
{
    public static void main(String[] args)
    {
        BankAccount b = new BankAccount(123456, "Sumit Mahato", 50000);

        b.display();
    }
}
```

Output:


java — -zsh — 161×46
sumit@Sumit java % date
Thu Jul 30 19:55:24 IST 2026
sumit@Sumit java % javac BankAccountDemo.java
sumit@Sumit java % java BankAccountDemo
Account Number : 123456
Account Holder : Sumit Mahato
Balance : 50000.0
sumit@Sumit java %

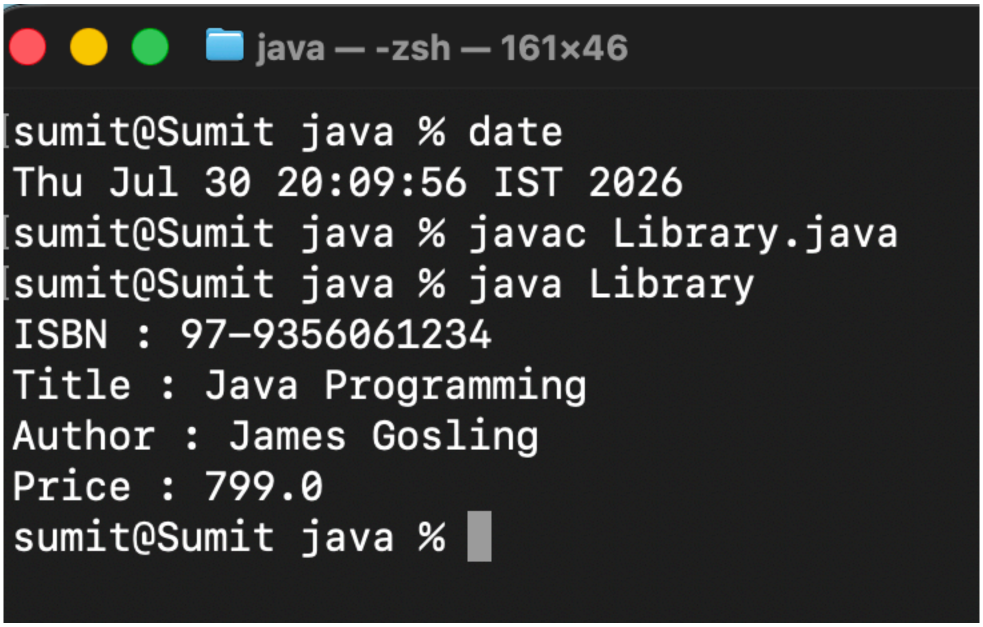
2.Create a **Library Book Management** program where the **book ISBN** is declared as **final** and cannot be changed once assigned. Display the book's ISBN, title, author, and price.

Source Code:

Users > sumit > Documents > java > Library.java > Library > main(String[])

```java
class Book
{
    final String isbn;
    String title;
    String author;
    double price;

    Book(String isbn, String title, String author, double price)
    {
        this.isbn = isbn;
        this.title = title;
        this.author = author;
        this.price = price;
    }

    void display()
    {
        System.out.println("ISBN : " + isbn);
        System.out.println("Title : " + title);
        System.out.println("Author : " + author);
        System.out.println("Price : " + price);
    }
}

public class Library
{
    Run | Debug
    public static void main(String[] args)
    {
        Book b = new Book("97-9356061234", "Java Programming", "James Gosling", 799);

        b.display();


    }
}
```

Output:



Conclusion:
The program successfully demonstrated the use of the final keyword in Java. A final variable cannot be reassigned after initialization, a final method cannot be overridden by a subclass, and a final class cannot be inherited. Thus, the final keyword helps protect data, prevents unwanted modifications, and improves the security and reliability of Java programs.